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The Interrelation Between Digital Competencies and Work-Life Balance Among Academic Staff – Realization of SDG Goals in the Science Sector

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Abstract

The ongoing digitalization of higher education generates the need for an in-depth analysis of how the adoption of digital technologies affects the functioning of sustainable academic staff. Despite the growing interest in work-life balance within sustainable development research, the relationship between the level of digital competences and employee well-being remains an area that requires further empirical investigation.

The aim of this study is to recognition the level of digital technology adoption among research and teaching staff at selected European higher education institutions, as well as to assess the impact of these technologies on the respondents' work-life balance. Quantitative research was used to achieve this objective. The research sample included eight universities. The participants were academic staff representing the fields of management and quality sciences, as well as economics and finance. The data collection instrument was a custom-designed diagnostic questionnaire, developed based on a review of contemporary frameworks concerning digital technology adoption and work-life balance. The data were analyzed using exploratory methods and machine learning algorithms within the JASP environment. The survey was voluntary and approved by the Rector's Committee for Ethics in Research Involving Human Subjects at the Hugon Kołłątaj University of Agriculture in Krakow.

The results of the study revealed that the level of professional digitalisation is multidimensional and does not result solely from technological skills, but also from one’s attitude towards private life and the way in which work-life balance is organised. Those who cope best with digitalisation effectively maintain close relationships, fulfil their private commitments and clearly set boundaries for their availability, being often supported in this by a strong motivation to be productive and/or a profound focus on relationships.

The project provided a foundation for developing a support model for academic staff and for formulating recommendations for university management policies in the context of the sustainable professional development of research and teaching personnel.

Introduction

The science sector plays a key role in creating and implementing sustainable development in the network of socio-economic relations. All sectors of the economy are currently undergoing a transformation towards sustainability and digitalisation, which presents them with new challenges. This convergence in the science sector is significant in three ways. Firstly, this sector is crucial as a generator of knowledge and a creator of trends in the Triple Helix model. Secondly, among the objectives of sustainable development, social aspects relating to the quality of life and work are just as important as environmental and business aspects. Finally, thirdly, the nature of academic work requires attention to work-life balance and the development of digital and social skills in order to effectively achieve professional goals, more so than in other professions.

The subject matter of this article concerns the issue of individual strategies of higher education staff regarding work-life balance and their relationship with the level of digital competences in the context of creating sustainable development policies in the academic community and universities as organisations. The importance of this issue is particularly emphasised in this profession.

The link between the digital competences of employees and aspects of work-life balance has been identified as a research gap, but not, as in most studies, by pointing to the fact that the better the competences of an employee, including digital competences, the better their work-life balance. In the study undertaken here, the objective was formulated as defining job crafting in relation to individual work and non-work priority strategies that determine the level of digital skills of academic staff.

In this context, two research questions were posed. Q1. What work–life balance (WLB) factors differentiate the level of occupational digitalization among university teachers? Q2. Which WLB-related attitudes and strategies – as measured by the Work–Nonwork Balance Crafting scale – are the most significant predictors of a high level of digital technology use in teaching and research activities?

The following stages of work were used to achieve this. Research methods typical for this type of study were used: critical analysis of the literature, analysis of primary data using a self-developed survey questionnaire (Likert Type Scale) and CART classification trees. The research sample comprised 245 researchers in selected European countries. Classification and Regression Trees were used in statistical analyses in the JASP programme. The dependent variable was the Synthetic Indicator of Occupational Digital Skills – a synthetic measure of digital technology adoption in academic teaching and research (Punie, Redecker 2017). In turn, the independent variable was developed based on 16 questions adopted from the Work–Nonwork Balance Crafting Scale (Kerksieck et al. 2022). The collected results were analysed in terms of previous research in the source literature; later, conclusions were proposed.

Literature Review & Theoretical Framework

The review of source literature took into account issues related to the implementation of SDGs at universities, the nature of academic work, the possibility of maintaining a work-life balance in the given profession in combination with the level of digital competences of the employees.

Sustainable development is the harmonious integration of economic, social and environmental goals. The Triple Helix model reinforces these processes by integrating various entities into a network of connections that allows for knowledge sharing, innovation implementation and improvement of the quality of life of the local community (Zadegan et al. 2025). The role of universities in this process is crucial due to their role in setting trends, conducting research and R&D cooperation with business and administration, educating specialists, acting as a catalyst, and promoting social responsibility values in cross-sector cooperation (Mêgnigbêto 2025, Zakaria et al. 2023). Their commitment is not limited to education, but extends to co-creating innovative solutions for socio-economic progress (Costa et al. 2021; Schebesch et al. 2024). They are generators of knowledge, centres for the integration of different communities, they create and bring together leaders of socially responsible and sustainable development (Filho et al. 2024).

When considering the role of the scientific community in achieving sustainable development goals, it is important to emphasise its impact on aspects such as SDG 3 Good Health and Well-being, SDG 4 Quality Education, SDG 8 Decent Work and Economic Growth, SDG 9 Industry, Innovation and SDG 11 Infrastructure and Sustainable Cities and Communities, as demonstrated by numerous studies, e.g.: Artyukhov (2024),

Galán-Muros et al. (2024), López-Santiago et al. (2024), Lumbreras & Moreno-Serna (2024), Pachava et al. (2025), Ferk Savec & Jedrinović (2025).

Higher education institutions, as entities of the science sector that are important for the development of sustainable communities, are also committed to implementing the SDGs within their organisations, which is in line with the subject matter of this article. In this context, it is worth noting The Times Higher Education University Impact Ranking 2024, which lists the top universities pursuing sustainable development goals and ranks 2,152 universities, (https://www.timeshighereducation.com/rankings/impact/overall/2024). Furthermore, the concept of social responsibility of universities has been introduced in the source literature (Popowska & Sady 2024; Zhu et al. 2024).

From the perspective of this paper, the most important aspects identified were the social aspects of universities as organisations, which include conditions conducive to development, equality and social justice, quality of life and social well-being, as well as physical and mental health. As workplaces, organisations should offer decent work and employment conditions, a safe working environment and opportunities for the development of social capital. In cross-sectoral cooperation between governments, businesses and civil society, universities are opinion-forming and influence the promotion of ethical values, diversity, a culture of tolerance and respect for human rights (Kwasek et al. 2025; Reena & Dinesh 2023). This also includes the issue discussed in this paper of education and the development of digital, social and civic competences, as well as the promotion of work-life balance and flexible forms of employment.

The nature of academics' work significantly determines the conduct of research on the essence of work-life balance in this profession. This refers to characteristics such as flexible working hours, self-management and self-organisation of work, the diverse nature of work, i.e. research, scientific, teaching and administrative, time pressure, high expectations and strong competition, which blur the boundaries between work and personal life. Diego-Medrano and Ramos Salazar (2021) highlight the excessive number of teaching duties, publication pressure and ever-growing administrative requirements, which contribute to longer working hours. Boamah et al. (2022) also point to a lack of support, stress and poor leadership. Hence, maintaining a work-life balance is particularly important for higher education staff.

Work-life balance is an interdisciplinary concept borrowed from psychology and applied to management sciences. Modern scientific definitions of work-life balance in the source literature were pioneered by Greenhaus, Beutell (1985), Clark (2000) and Frone (2003). The first scales for work-life balance mainly measured work-family conflict, evolving over time into a multidimensional approach that also includes role enrichment and other aspects of non-work life. Initial tools, such as the Greenhaus and Beutell scale (1985) and the Work Spillover Scale (Small, Riley 1990), serve as a reference point for later, more comprehensive measures of this phenomenon: Work-Family Spillover Scale (Grzywacz, Marks 2000), Work-Life Balance Self-Assessment Scale (Fisher 2001; Fisher-McAuley et al. 2003; Valcour 2007; Carlson et al. 2009; Yusuf 2018). Carlson et al. (2009) linked work–life balance with welfare, job satisfaction, and organisational commitment with family functioning. One of the most important points of reference for contemporary research on this issue is the paper by Casper et al. (2018), which is considered a coherent, modern approach to the study of work–life balance. The new definition of WNB according to Casper et al. (2018): “Employees’ evaluation of the favorability of their combination of work and nonwork roles, arising from the degree to which their

affective experiences and their perceived involvement and effectiveness in work and nonwork roles are commensurate with the value they attach to these roles" (p. 197).

Casper continued his research together with Wayne & Vaziri (2021), creating another tool comprising a total of 20 measurement items – 5 for global balance and 5 for each dimension. It was shown that global balance and its dimensions uniquely predict employee engagement, civic behaviour, organisational commitment, intent to leave, emotional exhaustion and/or health status, extending beyond existing measures of balance. This research therefore provides a comprehensive, validated, multidimensional tool for measuring work–non-work balance and offers a unique explanation of valued attitudes and behaviours.

The most recent work presenting the development and validation of a new scale for measuring work–nonwork balance, including the boundary management between the two spheres of life, is considered to be the article by Kerksieck et al. (2022). This study encompasses five countries (Austria, Finland, Germany, Japan, Switzerland) and confirms that active boundary crafting is a key mechanism for achieving subjective balance by adjusting one's style of functioning at work and outside of work. The results of the study suggest that job crafting strategies applied to work–nonwork balance management are universal in nature and can be used comparably in different cultural contexts. The scale developed by the authors is a tool that allows for better research and operationalisation of the processes of role boundary setting/blurring, and thus facilitates more precise research into the interdependencies between professional and non-professional life. The development of a multidimensional global measure of work–nonwork balance (global balance) allows for international comparisons and the examination of how work–life balance is linked with attitudes such as satisfaction, employee engagement, affectiveness (emotion) and effectiveness (efficiency) (Figure 1). This is one of the first tools measuring this construct at the international level, which was used in our own research described in this article.

Another aspect covered in this paper is digital competences and the use of AI in the work of academics. The role of higher education in harnessing the potential of AI to accelerate sustainable development is highlighted by the Higher Education Sustainability Initiative (HESI) (2024) Futures of Higher Education and Artificial Intelligence Action Group – Concept Note United Nations University, (https://sdgs.un.org/documents/futures-higher-education-and-artificial-intelligence-action-group-concept-note-56104). Digital competence in an academic context is defined as a complex set of knowledge, skills, and attitudes that enable the effective use of digital technologies in teaching, research, and administration. The European Digital Competence Framework for Educators (DigCompEdu) identifies six main areas: professional engagement, digital resources, teaching and learning, evaluation, student support, and developing students' digital competence (Punie, Redecker 2017). Research indicates that the digital competences of academic staff include not only technical skills, but also the ability to integrate technology into pedagogical practices and awareness of the ethical aspects of using digital tools. In the context of academic work, these competences extend to skills related to data management, communication of research results, and collaboration in international research teams.

Research on AI in higher education focuses mainly on technical aspects, academic integrity and teaching effectiveness, as well as the impact of AI on teaching and learning in academic environments (Artyukhov 2024; Kallunki 2024; Farrelly, Baker 2023; Vindaca et al. 2024). Pachava et al. (2025) introduce the AI

Academic Convergence (AIAC) Framework, demonstrating how generative AI can support personalized learning and operational efficiency. A very limited number of studies directly address the impact of AI on the work-life balance of academic staff, with only a few articles identified on this topic (Suresh, Kakkad 2023; AI in Education…, 2025). Research on AI for a sustainable society is in its initial stages, particularly in the context of higher education (Artyukhov 2024; Bong, Yunus 2025; Pachava et al. 2025; Ferk Savec, Jedrinović 2025).

It is worth noting that, in view of the growing possibilities for the use of AI in the professional activities of academics, work-life balance deserves a new look in the context of the concept of job crafting. Wrzesniewski & Dutton (2001) emphasized that employees are not merely passive performers of assigned tasks, but actively transform their work to give it more meaning, improve the quality of their professional life, and better adapt it to their own expectations. Defined as a process in which employees proactively modify the tasks, relationships, and/or cognitive aspects of their work to better suit their needs, values, and capabilities (Berg et al. 2008; Cárdenas-Muñoz, Campos-Blázquez 2023), it takes on new meaning when considering the possibilities of incorporating AI tools into their research and teaching work.

Although the source literature contains studies on individual aspects of the issues discussed, there is a need for comprehensive research linking the digital competences of academic staff with the achievement of specific SDGs in the context of work-life balance (Simanullang et al. 2024). No studies have been found in the published literature that combine individual strategies for work-life balance with digital skills in specific professional groups, especially in the academic environment. This area of our research was defined as a research gap and is considered an innovative contribution to the source literature.

Methodology

In order to answer the following two research questions:

Q1. *What work–life balance (WLB) factors differentiate the level of occupational digitalization among university teachers?*

Q2. *Which WLB-related attitudes and strategies – as measured by the Work–Nonwork Balance Crafting scale – are the most significant predictors of a high level of digital technology use in teaching and research activities?*

the results of a survey conducted in six countries among 245 research and teaching staff employed at eight universities were used. The respondents represented the following disciplines: management and quality sciences, economics, and finance. The survey was voluntary and approved by the Rector's Committee for Ethics in Research Involving Human Subjects at the Hugon Kołłątaj University of Agriculture in Krakow. All participants were adults and were informed in advance about the purpose and rules of participation in the study.

Initially, 245 academic and research staff employed at eight higher education institutions participated in the study. Three of these institutions were based in Poland, while the remaining five were foreign academic institutions located in Spain, Portugal, the Czech Republic, Moldova, and Ukraine. After analyzing the

consistency of the data set and removing outliers, the research team qualified 191 respondents for the final sample.

The largest group of respondents were academic staff representing Poland, whose share in the total sample was approximately 37.2%. This was followed by respondents from the Czech Republic (approximately 18.8%), Spain (approximately 17.8%), Ukraine (approximately 12%), and Portugal (approximately 10.5%). The least represented group were respondents from Moldova, who accounted for approximately 3.7% of the total number of participants.

Women accounted for the majority of respondents (59.2%). Men accounted for approximately 40.8% of the total number of respondents.

The largest age group in the sample was people aged 45–49, who accounted for approximately 24.6% of the sample. Respondents aged 50–54 (approximately 18.8%), 35–39 (approximately 13.6%), and 40–44 (approximately 12.6%) also constituted a significant proportion. The smallest percentage was represented by the oldest and youngest respondents (Fig. 2).

The largest segment of the research sample was represented by people with at least 16 years of professional experience in higher education. Their share was approximately 49.2% of the total research sample. The smallest percentage were people who had been working at a university for one year (approximately 7.3%) (Fig. 3).

In terms of academic position, the most frequently indicated was "research and teaching assistant professor" — this position was held by approximately 39.8% of all respondents. Detailed information on the positions held by the respondents is presented in Fig. 4.

The respondent profile also included statements regarding net monthly income, expressed in euros. Nearly one-third of respondents reported incomes ranging from €1,000 to €1,700. Another large group consisted of respondents earning between €1,700 and €2,400 per month, accounting for approximately 30.4% of the total. Detailed data in this regard is presented in Fig. 5.

Looking at the structure of respondents' place of residence, it can be observed that the largest percentage of them lived in very large cities with over 500,000 inhabitants - this group accounted for approximately 37.2% of all respondents. Approximately 27.2% of respondents lived in large cities (100,000–499,999 inhabitants), and approximately 16.8% in medium-sized cities (20,000–99,999 inhabitants). A total of about 15.2% of respondents lived in rural areas, including about 7.9% in towns with fewer than 2,000 inhabitants and about 7.3% in larger towns. The smallest group were respondents living in small towns (5,000–19,999 inhabitants), whose share was approximately 3.7% (Fig. 6).

The study used the classification and regression trees (CART) to identify factors differentiating the level of digital advancement of academic teachers employed at universities. The analysis aimed to determine the impact of selected predictors on the value of the dependent variable, which was a synthetic indicator of professional digitalization. This indicator served as a measure of the overall level of adaptation of digital technologies in teaching and research activities.

The independent variables included individual questions from the standardized Work–Nonwork Balance Crafting Scale (Kerksieck et al. 2022). This scale allows us to assess specific strategies and actions taken by employees in order to shape and maintain a balance between their professional and non-professional lives. The study used a version of the tool adapted to the specific language of the research sample.

For the analysis using the classification and regression trees (CART), 181 respondents were finally selected from among the total of 191 respondents. Ten participants who had the highest level of digital skills and differed significantly from the rest of the research group were excluded. Due to the specific nature of this subgroup, a separate section of the analysis was devoted to it.

The C&RT method was chosen for data analysis because of its usefulness in analyzing nonlinear relationships and its ability to identify key subgroups that differ in the level of the dependent variable. Regression models based on decision trees made it possible to identify the most significant predictors of digitalization and to indicate combinations of characteristics conducive to a higher level of digital technology adaptation.

The basis for the construction of the synthetic variable of professional digitalization was data obtained from a questionnaire in which respondents declared their assessment of the frequency of use of individual digital tools in teaching and research. Responses were given on a five-point Likert scale, where 1 meant no use of a given tool and 5 meant regular and systematic use in professional practice.

The synthetic variable was constructed on the assumption that the effective functioning of a research and teaching employee in the realities of contemporary higher education requires the parallel development of digital competences in both of these domains. Therefore, the sub-variables related to teaching digitalization and research digitalization were given equal weight in the final model.

In order to maintain the appropriate sensitivity of the measure, not only the frequency of use but also the level of technological advancement of a given tool was taken into account. The level of advancement of each digital tool was assigned based on the skill levels defined in the European Digital Competence Framework for Educators (DigCompEdu) (Punie & Redecker 2017). These levels were assigned to the tools to reflect the competencies of the employees using them. Next, each tool was assigned appropriate weight values according to the following table: A1 – novice (weight 1), A2 – explorer (weight 2), B1 – integrator (weight 3), B2 – expert (weight 4), C1 – leader (weight 5) and C2 – pioneer (weight 6) (Punie & Redecker 2017).

Thus defined weights allowed for differentiation of the impact of individual tools on the final form of the variable, in line with the assumption that the use of more technologically complex tools requiring greater competence should be rewarded more strongly than the use of technologies that are relatively easy to use.

The weights were assigned separately for variables related to teaching and research work. The final division is presented in the two tables below (Table 1 and Table 2).

Table 1. Classification of tools used in educational work

Full name of educational variable	DigCompEdu level	Weight
Office software	A1 – Novice	1
Video conferencing tools	A2 – Explorer	2
AI models	C1 – Leader	5
VR technologies	C1 – Leader	5
E-learning platforms	B1 – Integrator	3
Social media	A2 – Explorer	2
Cloud services	A2 – Explorer	2
Multimedia resources	A1 – Novice	1
Grading tools	B1 – Integrator	3
Presentations and dynamic tools	A2 – Explorer	2
Virtual laboratories and simulations	B2 – Expert	4
Online survey software	B1 – Integrator	3
Interactive tools	B2 – Expert	4
Group management tools	B2 – Expert	4
Note-taking and knowledge management tools	B1 – Integrator	3
Self-reflection tools	B2 – Expert	4
Quiz creation tools	B1 – Integrator	3
Plagiarism detection tools	B1 – Integrator	3
Tools for checking the use of AI in texts	C1 – Leader	5

Source: own study in 2025

Table 2. Classification of tools used in research work

Full name of the research variable	DigCompEdu level	Weight
Office software	A1 – Novice	1
Video conferencing tools	A2 – Explorer	2
AI models	C1 – Leader	5
VR technologies	C1 – Leader	5
Collaboration tools	B2 – Expert	4
Tools for scientific communication	B1 – Integrator	3
Video conferencing tools	A2 – Explorer	2
Repositories and databases	B2 – Expert	4
Social media	A2 – Explorer	2
Cloud services	A2 – Explorer	2
Multimedia resources	A1 – Novice	1
Data analysis and visualization	C1 – Leader	5
Presentations and dynamic tools	A2 – Explorer	2
Tools for text and qualitative data analysis	B2 – Expert	4
Virtual laboratories and simulations	B2 – Expert	4
Online survey software	B1 – Integrator	3
Text editors supporting scientific writing	B2 – Expert	4
Tools for managing scientific publications	B1 – Integrator	3
Tools for managing research projects	B2 – Expert	4
Tools for organizing meetings and schedules	A2 – Explorer	2
Tools for literature review	C1 – Leader	5
Plagiarism detection tools	B1 – Integrator	3
Tools for checking the use of AI in texts	C1 – Leader	5
Time management and productivity tools	A2 – Explorer	2
Note-taking and knowledge management tools	B1 – Integrator	3
Image and video analysis tools	C1 – Leader	5
Big data analytics and machine learning tools	C2 – Pioneer	6
Self-reflection tools	B2 – Expert	4

Source: own study in 2025

After assigning each digital tool an appropriate level of technological advancement along with a corresponding weighting value, it was possible to calculate quantitative variables describing their digital usage. The analysis used two partial quantitative variables:

- didactic digitalization – a synthetic indicator of the level of advancement in the use of digital tools in the field of teaching;
- research digitalization – an analogous indicator referring to tools used in scientific and research activities.

These indicators were constructed based on the assumption that each tool should influence the final result in proportion to two factors: the frequency of its use (as declared by the respondent) and the technological advancement of the tool (expressed by an appropriate weight).

For each respondent, the value of the following products was determined:

$$O_{i,j} * W_{i,j} \qquad (1)$$

where:

Oi,j – respondent's response value for the i-th teaching tool or j-th research tool (on a scale of 1–5),

Wi,j – assigned weight for the i-th teaching tool or j-th research tool (on a scale of 1–6)

Next, the sum of these products was divided by the sum of the weights for all the tools assessed in a given area, according to the following formulas:Oi, j – respondent's response value for the i-th teaching tool or j-th research tool (on a scale of 1–6),

$$Didactic\ digitalization = \frac{\sum_{i=1}^{n} O_i * W_i}{\sum_{i=1}^{n} W_i} \qquad (2)$$

$$Research\ digitalization = \frac{\sum_{j=1}^{m} O_j * W_j}{\sum_{i=j}^{m} W_j} \qquad (3)$$

where:

n – number of teaching variables (in this study: 19),

m – number of research variables (in this study: 28)

The partial variables calculated in this way were continuous and took values from 1 to 5, with higher values indicating both greater intensity of digital technology use and a higher level of advancement.

The final variable, called "professional digitalization," was determined as the weighted average of the two components. In accordance with the previously adopted assumption of the equivalence of teaching and

research functions in the work of a research and teaching employee. Each component was assigned a weight of 0.5. The final formula took the following form:

$$Professional\ digitalization = 0.5 * Didactic\ digitalization + 0{,}5\ * Research\ digitalization \quad (4)$$

The variable constructed in this way was used in further quantitative analysis as an indicator of the overall level of digital competence of academic teachers (dependent variable). It is worth noting that this variable not only integrates the frequency of use of various tools, but also takes into account their diverse technological and innovative potential. This makes it a much more sensitive and cognitively valuable measure than the classic averaging of responses on the Likert scale.

The range of the variable (from 1 to 5) has been logically divided into six adapted interpretation intervals that determine the level of use of new technologies by users, in line with the progression of the DigCompEdu framework (Punie & Redecker 2017) (Table 3).

Table 3. DigCompEdu level breakdown for this study.

Level	Scope
A1 – Novice	1.00 – 1.66
A2 – Explorer	1.67 – 2.33
B1 – Integrator	2.34 – 3.00
B2 – Expert	3.01 – 3.67
C1 – Leader	3.68 – 4.34
C2 – Pioneer	4.35 – 5.00

Source: own study.

Results

The C&RT regression decision tree model was constructed based on data from 181 respondents. Its purpose was to identify the most important factors differentiating the level of the dependent variable (professional digitalization) in the context of the variables describing it (work-life balance).

The regression model used 16 independent variables corresponding to the statements of the Work-Nonwork Balance Crafting Scale (Kerksieck et al. 2022): V1 – If I must get personal chores done during working time, I make sure that my work will not be negatively affected.; V2 – When I must get some work chores done, I come home later or go to work earlier, if necessary; V3 – In some situations, I temporarily emphasize my work (e.g., work more before vacations to get things done); V4 – In certain phases of my life, I temporarily prioritize my work life to achieve a work goal; V5 – I try hard to meet my professional obligations, even if I'm demanded strongly by my private life; V6 – When I'm in a bad mood because of personal matters, I try not to

let this affect my work environment; V7 – I make sure that I can enjoy the pleasant aspects of my work, even though I'm strongly demanded by my private life; V8 – I tell people of my private environment when I'm unable to communicate with them during working time or to take care of private matters; V9 – If I must get work chores done during leisure time, I make sure that my personal life will not be negatively affected; V10 – When I must get some personal chores done, I come to work later or go home earlier, if necessary; V11 – In some situations, I temporarily emphasize my private life (e.g., when a friend needs my support); V12 – In certain phases of my life, I temporarily prioritize my private life to achieve a nonwork goal; V13 – I try hard to meet my private obligations, even if I'm demanded strongly by my work; V14 – When I'm in a bad mood because of work matters, I try not to let this affect my personal environment; V15 – I make sure that I can enjoy the time with my partner, my family or my friends even though I'm strongly demanded by my work; V16 – I tell people of my professional environment when I'm unable to communicate with them during leisure time or to take care of professional matters.

A complexity penalty parameter of 0.030 was used to construct the decision tree. During the construction process, 78 divisions were made, which allowed for a detailed differentiation of subgroups in the training set.

The analysis was performed using the JASP program on a total of 181 observations. Of these, 154 cases (approximately 85%) were used to train the tree, while 27 observations (approximately 15%) were used to evaluate the quality of the predictions.

In order to assess the accuracy of the regression decision tree used to predict the level of digital literacy of academic teachers, an analysis of prediction error indicators was carried out. Table 4 presents five commonly used measures of regression model quality that allow for the assessment of the accuracy and stability of the obtained predictions.

Table 4
Decision tree model quality indicators

Indicator	Value
MSE	0.181
MSE (scaled)	0.689
RMSE	0.425
MAE / MAD	0.381
MAPE	14,63%

Source: own study based on the results of an analysis performed with the JASP program (2025).

The MSE (Mean Squared Error) indicator, which in the analyzed model took the value of 0.181, indicated a moderate average discrepancy between the predicted and actual levels of professional digitalization. Its scaled version (MSE scaled), which was 0.689, suggested that the model explains a significant part of the

observed variance while maintaining a moderate level of error in relation to the total variation of the dependent variable in the sample.

The RMSE (Root Mean Squared Error) indicator, equal to 0.425, and MAE / MAD (Mean Absolute Error / Median Absolute Deviation), with a value of 0.381, indicate that the average deviation of the prediction from the empirical values remained at a level not exceeding 0.4 points. These results indicate moderate but stable accuracy of the regression model. On the other hand, the MAPE (Mean Absolute Percentage Error) indicator, at 14.63%, means that the average forecast error was approximately 14.6% of the actual values. In the context of social research, this level of accuracy can be considered relatively satisfactory.

The metric of the average decrease in the loss function was used to assess the relative weights of the independent variables (Table 5).

Table 5
Significance of variables in the regression model (C&RT)

Variable	Relative Importance	Mean Dropout Loss
V15	19.504	0.543
V13	12.946	0.568
V2	11.084	0.477
V11	10.375	0.571
V16	8.693	0.480
V3	7.987	0.475
V12	7.897	0.450
V14	7.170	0.477
V7	3.574	0.450
V8	2.668	0.450
V4	2.428	0.450
V5	2.178	0.450
V6	1.981	0.450
V10	1.222	0.450
V1	0.450	0.450
V9	0.293	0.450

Source: own study based on the analysis performed with the JASP program (2025).

The analysis highlighted five variables of the highest significance. One of the key predictors was the variable relating to concern for time with loved ones, even in conditions of professional overload (V15). Its relative

value was 19.504, and the average decrease in model accuracy after its exclusion reached 0.543. This indicated that concern for private relationships in the context of work overload strongly differentiated the level of professional digitalization.

The response concerning the importance of fulfilling private commitments despite pressure from work (V13) also achieved high significance. The importance index for this variable was 12.946, and the accompanying dropout loss reached 0.568. This result indicated a strong link between consistent fulfillment of personal commitments and developed digital competencies.

The third most influential factor in the model was the strategy of flexibly adjusting working hours to professional duties (V2). The significance value of this variable was 11.084, which may suggest that people who demonstrate such flexibility are better at adapting to digital tools that support work organization.

The variable concerning temporarily prioritizing private life in specific situations (V11) received a value of 10.375 with an average dropout loss index of 0.571. These results indicate the important role of life flexibility in the context of developing professional digital skills.

The declaration of the need to inform others in advance about limited availability in professional or private situations (V16) was also important for the construction of the model. This variable achieved a value of 8.693 and a dropout loss rate of 0.480, which emphasizes the importance of consciously communicating boundaries as a strategy conducive to the efficient use of technology.

The resulting tree (Fig. 7) has a hierarchical structure in which divisions based on the values of independent variables are analyzed in sequence. Each end node of the tree represents a subgroup of respondents with a similar level of professional digitalization, which allows for the identification of key predictors and their decision thresholds.

The regression tree created based on the C&RT method was rooted in variable V15, which concerned the following statement: “I make sure that I can enjoy the time with my partner, my family or my friends even though I’m strongly demanded by my work.” The first division in the model structure was made based on the value of this variable, using a decision threshold of 1.5. Respondents who indicated a value below 1.5, i.e., answered “1 – strongly disagree,” were classified to the left branch. For this subgroup, the model predicted an average level of professional digitalization of 1.92. This value, close to the lower limit of the five-point scale (1–5), indicated a low level of advancement in the use of digital technologies in teaching and research.

After the first fork, in which respondents were divided according to the value of variable V15, those who assigned a value equal to or higher than 1.5 to this statement were classified to the right branch of the tree. This group, constituting the vast majority of respondents, was then divided based on variable V13, which concerned the following statement: “I try hard to meet my private obligations, even if I’m demanded strongly by my work.” Those who assigned a value lower than 1.5 to variable V13 (strongly disagreed with this statement) were assigned to a small subgroup n = 5, which was further divided based on variable V16. In this part of the tree, the model analyzed responses to the statement: “I tell people of my professional environment when I’m unable to communicate with them during leisure time or to take care of professional

matters.” The responses to this statement allowed us to identify two final nodes. For the four respondents (n = 4) who assigned a value below 4.5 to this statement, the model estimated the average level of professional digitalization at 1.72. This result, close to the lower limit of the scale, indicates a very low level of digital advancement. These respondents not only rejected the need to systematically fulfill their private commitments (V13 < 1.5), but also did not show full readiness to actively communicate their own availability boundaries.

A single case (n = 1) presented a different characteristic, in which the respondent declared high agreement with statement V16 (value ≥ 4.5). In this case, the model assigned a professional digitality value of 3.20, which is a moderate result, significantly higher than in the other nodes of this branch.

After passing through subsequent forks in the tree, starting with variable V15, the model classified respondents with at least a minimum level of agreement with this statement (V15 ≥ 1.5) for further analysis. Then, within this group, variable V13, which refers to the willingness to fulfill private commitments despite professional overload, played a key role. Those who agreed with this statement at least to a minimal extent (V13 ≥ 1.5) were further divided based on variable V11 (“In some situations, I temporarily emphasize my private life (e.g., when a friend needs my support”). For respondents who assigned a value of 2.5 or higher to V11, the next branch was again based on variable V13. This time, however, the decision threshold was 4.5, which means that a further division was made between those who strongly agreed with this statement (value 5) and the rest. The final node covered by this analysis includes a group of n = 37 respondents who expressed very high agreement with statement V13, assigning it a value equal to or higher than 4.5. These are people who declare with great conviction that they consistently fulfil their private obligations despite their professional burdens. The model assigned this group an average level of professional digitalization of 2.27, which is slightly below the middle of the scale.

The next part of the tree includes respondents who assigned a value equal to or higher than 1.5 to variable V15 in the first branch. The respondents were then classified into a branch characterized by at least moderate attachment to private life, in accordance with the values of variables V13 and V11 meeting the criteria V13 ≥ 1.5 and V11 ≥ 2.5. A further division was made for variable V13, this time with a threshold value of 4.5. Respondents who did not give this statement the maximum value were assigned to group n = 83. Within this group, a further division was made based on variable V15 with a decision threshold of 4.5. Respondents who declared moderate agreement with statement V15 (below 4.5) were further divided according to their responses to variable V2, i.e.: „When I must get some work chores done, I come home later or go to work earlier, if necessary.” For most respondents in this branch (n = 60) who indicated a value greater than or equal to 2.5 for variable V2, the model assigned a professional digitality value of 2.46. This result falls within the lower range of the scale and indicates a relatively low level of digital advancement among the respondents. In contrast, four people (n = 4) who assigned a value below 2.5 to variable V2 obtained a predicted professional digitality value of 3.18. In this case, the model indicated a significantly higher level of digital advancement among the respondents.

The next leaf of the decision tree identifies a group of nineteen respondents whose predicted level of professional digitalization was 2.84 on a five-point scale. To be included in this subgroup, the respondents had to meet a sequence of conditions based on four statements. First, the model retained only those who

agreed at least to a minimal extent with the statement that they are able to enjoy time with their loved ones despite work overload (V15 ≥ 1.5). In the next step, those respondents who declared at least moderate consistency in fulfilling their private commitments despite work pressure (V13 ≥ 1.5) were retained. Third, the algorithm selected individuals who were willing to temporarily prioritize their private life when the situation required it (V11 ≥ 2.5). The fourth condition again concerned variable V13, but this time the model retained only those respondents who did not give the maximum score for this statement (V13 < 4.5), indicating a certain restraint in declaring absolute readiness to make professional sacrifices for the sake of their personal lives. The last filter again referred to variable V15: within the group defined in this way, those who gave it the highest possible score (V15 ≥ 4.5) were selected, meaning that they strongly emphasized the value of relationships with loved ones.

The next part of the tree structure analysed included the responses of respondents who, in the first branch, assigned a value equal to or higher than 1.5 to variable V15 (“I make sure that I can enjoy the time with my partner, my family or my friends even though I’m strongly demanded by my work”). Next, respondents who declared at least minimal consistency in fulfilling their private responsibilities despite being overworked (V13 ≥ 1.5) and those who rated their willingness to temporarily prioritize their private life below the threshold of 2.5 (V11 < 2.5) were included. Within this group, made up of a total of 23 people, a further division was made based on the value of variable V15. This time, however, a new threshold was set – 2.5. For eight respondents who rated V15 below 2.5, i.e., showed little concern for their personal life despite their previously declared flexibility in the private sphere, the model applied the V3 variable. Variable V3 concerned the statement: “In some situations, I temporarily emphasize my work (e.g., work more before vacations to get things done).” The responses of eight people made it possible to identify two final nodes For five respondents (n = 5) who rated statement V3 below 4.5, i.e., did not explicitly declare a high willingness to temporarily intensify their work, the model assigned a professional digitalization value of 2.14. This result falls within the lower range of the scale and suggests a limited level of adaptation to digital technologies among these respondents. A different profile was represented by three respondents (n = 3) who rated V3 at 4.5 or higher, thus showing a high readiness to temporarily increase the intensity of their professional activities. In their case, the predicted level of professional digitalization was 3.10, which is above the average level.

The final branch of the regression tree included individuals who were classified as respondents with a low level of readiness to prioritize their private life (V11 < 2.5), while showing a moderate level of concern for relationships with loved ones (V15 ≥ 2.5). Within this group (n = 23), a further division was made based on the responses to statement V14: “I make sure that I can enjoy the time with my partner, my family or my friends even though I’m strongly demanded by my work.” Although variable V14 is very similar in substance to V15, it was treated as an independent source of information, which indicates the need to capture the repeatability of the respondents' statements regarding their personal relationships. The decision threshold was set at 3.5, which corresponds to the separation of those expressing moderate agreement from those who declared strong agreement. The first of the final nodes included five respondents (n = 5) who rated V14 below 3.5 – they did not show a clear commitment to personal relationships. The model assigned this group a predicted professional digitalization value of 2.50, which falls within the lower range of average scores and suggests limited integration of digital technologies in the respondents' professional work. The second

terminal node included ten respondents (n = 10) who assigned a value equal to or higher than 3.5 to statement V14, thus agreeing to a significant extent with the idea of actively caring for relationships with family and friends. For this group, the model estimated the level of professional digitalization at 3.23, which is one of the highest values in the entire tree.

Analysis of responses from respondents with the highest levels of professional digitalization

Based on the average responses of respondents with the highest digital skills (average professional digitalization = 4.1), a clear pattern can be observed, with high values assigned to most variables of the Work–Nonwork Balance Crafting scale. Particularly high ratings were given to statements V1–V5, which refer to the respondents' ability to maintain professional effectiveness despite the need to perform personal duties, flexibly adjust their working hours, and temporarily intensify their professional efforts. This indicates a strong ability of the respondents to strategically manage their work rhythm (Table 6).

Table 6
Average response values for the Work–Nonwork Balance Crafting scale in the group of respondents with the highest level of professional digitalization.

Variable no.	On average
V1	4.6
V2	4.1
V3	4.6
V4	4.3
V5	4.5
V6	4.3
V7	4.2
V8	3.3
V9	3.5
V10	3.3
V11	4.3
V12	4.2
V13	3.2
V14	3.9
V15	4.5
V16	3.7

Source: own study.

The respondents also showed a high willingness to protect their own mental welfare and maintain emotional balance (e.g., V5 = 4.5, V6 = 4.3). Relatively lower, though still moderately high, average values were found in the area of consistent fulfillment of private responsibilities (V13 = 3.2) and informing others about their unavailability (V16 = 3.7). This may suggest that even digitally advanced individuals do not always demonstrate full assertiveness and organization in protecting their personal time.

It is worth noting the significantly lower average values for variables V8–V10, which describe operational adjustments during work or private time. This demonstrates that even highly digital individuals may prefer more integrated strategies that do not require frequent reorganization of their work schedule. The profile of these respondents shows work–life balance as a consciously shaped process based on a high sense of control, consistency, and strategic management of time and energy.

Discussion

The analysis undertaken in this paper fills a research gap and brings a new, comprehensive perspective to the source literature in the field of management, taking into account the multifaceted nature of digitalization and psychosocial and organizational factors. Most of the research published to date links work-life balance to the concepts of job satisfaction and work efficiency, while empirical long-term studies on the impact of AI on the welfare of academic staff are still in their early stages (Kallunki et al. 2024), although they appear to be extremely important for sustainable development in the future. The study discusses these relationships as the implementation of SDG goals in the science sector.

The results of the study indicate that the level of digital competence of academic staff significantly correlates with the quality of work-life balance, which is consistent with the observations of Garini and Muafi (2023), Mendoza Velazco, (2024) – better mastery of digital tools allows for more effective performance of professional tasks, which can lead to time savings. In turn, the link between work-life balance as an important factor influencing the welfare and satisfaction of academic employees is confirmed by the findings of Diego-Medrano and Ramos Salazar (2021).

The issue of choosing job-crafting working and non-working aspects examined here can be compared with the findings of Duan and Deng (2024), who identified the psychological need for achievement and work-life balance as important factors influencing work efficiency. These authors also demonstrated that the need for autonomy indirectly affects work efficiency—its impact is fully mediated by work–life balance. This can be compared to the choice of cognitive/emotional, physical, or relational work–life balance strategies according to the measures used in the authors' study.

Studies conducted among researchers in Serbia and Australia are particularly noteworthy. In the case of Serbia, Vukelić et al. (2021) point to the phenomenon of job crafting as gaining particular importance in the context of increasing digitalization and AI. This is consistent with the results obtained in this study, which show that the strategy of consciously shaping one's own work can be an effective tool for managing work-life balance.

The case of Australia (Miranda, Khan 2022), on the other hand, highlights the complexity of digital professional competencies and their impact on work-life balance, but not from such a multidimensional psychosocial perspective as in the analysis described in this paper.

Nadapdap et al. (2025) reached very similar conclusions. According to these authors, the digitization of academic staff should not be limited to the development of technical skills, but should also integrate issues of work organization and mental health, in line with the conclusions drawn in the study described here. The organizational barriers to the implementation of digital technologies observed in the study point to the need to develop new human resource policies in the science sector that combine the development of digital competences with a concern for work-life balance, as also highlighted by Pradita and Franksiska (2020). The importance of institutional support and human resource management in science is consistent with the literature on the implementation of HR strategies in the research sector (Suresh, Kakkad 2023). Furthermore, the use of generative AI in higher education presents new challenges and opportunities for scientific

institutions, requiring a systemic approach and sustainable staff development, in line with similar recommendations by Pachava et al. (2025).

The link between professional digitalization and individual work-life balance strategies seems to be innovative in the described project. Job crafting strategies—understood as active, multifaceted shaping of one's work model and private sphere—determine the degree of digital advancement in the academic environment. This is an innovative combination of organizational psychology and research on the digitization of academic staff. The relationship between digitalization and work-life balance is assessed here on the basis of original questionnaire measures. The study may serve as a starting point for further, long-term monitoring of the impact of AI and digitization on work-life balance and the sustainable development of academic staff.

Conclusions

The regression tree analysis allowed us to identify several key configurations of attitudes and strategies in the area of work–life balance, which differentiated the level of professional digitalization of the respondents. In response to the first research question, the strongest dividing factor was variable V15 (“I make sure that I can enjoy the time with my partner, my family or my friends even though I’m strongly demanded by my work”). This was the starting point for the entire decision-making structure. Any disagreement with this statement (V15 < 1.5) resulted in the respondent being clearly assigned to the group with a very low level of professional digitalization. This emphasized the importance of relational orientation as a fundamental component of the model. The results are illustrated in figure 8.

Further divisions showed that consistency in fulfilling private commitments despite professional pressure (V13) was the second key factor differentiating the respondents. Respondents who did not demonstrate such consistency (V13 < 1.5) achieved low or very low levels of digitalization—with the exception of those who nevertheless declared their willingness to communicate boundaries (V16 ≥ 4.5), which acted as an important compensating factor. High awareness of the need to regulate professional availability proved to be an independent predictor of professional digitalization, regardless of other aspects of the WLB.

In further branches of the decision tree, the model indicated that even a very strong orientation towards private life (V13 ≥ 4.5) did not guarantee a high level of professional digitalization among the respondents if it was not supported by other integrated strategies. For example, individuals who showed high concern for relationships (V15 ≥ 4.5) but did not assign the maximum value to variable V13 achieved moderate levels of digitalization (2.84), suggesting the importance of a multidimensional approach to work-life balance. On the other hand, the model showed that people who declared a low orientation towards private life (V11 < 2.5) but at the same time showed a willingness to temporarily intensify their work (V3 ≥ 4.5) were characterized by higher professional digitalization (3.10). This meant that components related to productivity and professional ambition could also contribute to the digital adaptation of the respondents.

The highest digitalization score (3.23) was obtained in the group of respondents who, despite low readiness to reorganize their priorities (V11 < 2.5), consistently declared strong commitment to their private

relationships (V14 ≥ 3.5). This result suggests that a clear relational orientation can play a stabilizing and strengthening role, even in the context of deficits in other areas of the WLB.

In response to the second question posed in the study, it was found that the professional digitalization of respondents was not the result of a single attitude or declaration, but resulted from patterns of coexisting relational, boundary and productive strategies (figure 9). The key aspects in this regard were: consistency in the respondents' private activities (V13), the ability to communicate their boundaries (V16), the intensification of the respondents' professional activities at selected moments (V3), and their genuine concern for relationships (V15 and V14).

These results confirm that the adaptation of digital technologies in the academic environment is systemic and closely linked to the broadly understood management of work-life balance. The resulting model shows that the level of professional digitalization did not result solely from technological skills, but was strongly related to attitudes towards private life and the way in which the respondents organized the balance between their professional and personal lives. The highest professional digitalization scores were achieved by people who were able to simultaneously maintain relationships with their loved ones, consistently fulfill their private commitments, and clearly set boundaries for their availability. In some cases, the score was influenced by a strong motivation to be productive or a deep relational orientation. Professional digitalization therefore appears to be a multidimensional phenomenon, the full determinants of which go beyond a simple attribution to the level of technological knowledge.

Based on the results of the regression model, specific recommendations can be formulated for higher education institutions and organizations employing academic and teaching staff. First and foremost, it should be emphasized that the development of employees' digital competences should not be viewed solely as the result of technological training, but as a phenomenon strongly linked to attitudes towards work-life balance. People who are able to protect their relationships with their loved ones, manage their time flexibly, and consciously set boundaries between their professional and personal lives achieved a significantly higher level of professional digitalization. From a human resource management perspective, this means that supporting work-life balance strategies should be an integral part of a university's digitalization policy. Institutions that enable flexible forms of work, promote a culture of mutual respect for time limits, and invest in the welfare of their employees create an environment conducive to technological adaptation. Moreover, the data indicate that even moderate concern for privacy— as long as it is consistent and communicated clearly—can significantly support the development of digital skills.

In light of these findings, it is recommended that efforts to increase the level of digitization in institutions not be limited to technical aspects. It is equally important to implement organizational solutions that support employee autonomy, their ability to recharge, and long-term energy management. As the model structure shows, these factors form the foundation for effective functioning in a highly digital work environment.

Limitations & Future Research

Among the limitations, we can definitely mention the significantly smaller research group in one of the countries surveyed – Moldova (3.7%). In the next stages of the research, a decision will be made on whether to increase the research sample or exclude Moldova altogether, considering the logistical difficulties in collecting data from this country. The authors will continue their research in the context of the connection between career paths and financial situations and how scientists develop their digital skills, using an organizational or individual approach. The study participants may be included in a long-term analysis of the impact of digitization and AI tools on the work of academic teachers.

Declarations

Author Contribution

M.N., A.N., Z.G.-S., K.P. and B.P. contributed to conceptualization and methodology. M.N. contributed to software. M.N., A.N., Z.G.-S. and D.A. contributed to validation. M.N. and D.A. contributed to formal analysis. M.N., A.N., Z.G.-S. and D.A. contributed to investigation. M.N., A.N. and Z.G.-S. contributed to data curation. M.N., K.P., D.A., V.T., F.P. and B.P. contributed to resources. M.N., A.N., Z.G.-S. and D.A. contributed to writing & original draft preparation. M.N., A.N., Z.G.-S., D.A., V.T., F.P. and B.P. contributed to writing, review & editing. M.N., D.A. and B.P. contributed to visualization. M.N. and D.A. contributed to supervision. M.N. contributed to project administration. M.N. contributed to funding acquisition. All authors reviewed and approved the final manuscript.

Data Availability

The data were obtained from primary research conducted in accordance with ethical principles. The article is co-financed by the Minister of Science under the ‘Regional Initiative of Excellence’ programme. Agreement No. RID/SP/0039/2024/01. Subsidised amount PLN 6,187,000.00. Project period 2024–2027.The survey was voluntary and approved by the Rector's Committee for Ethics in Research Involving Human Subjects at the Hugon Kołłątaj University of Agriculture in Krakow.

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Figures

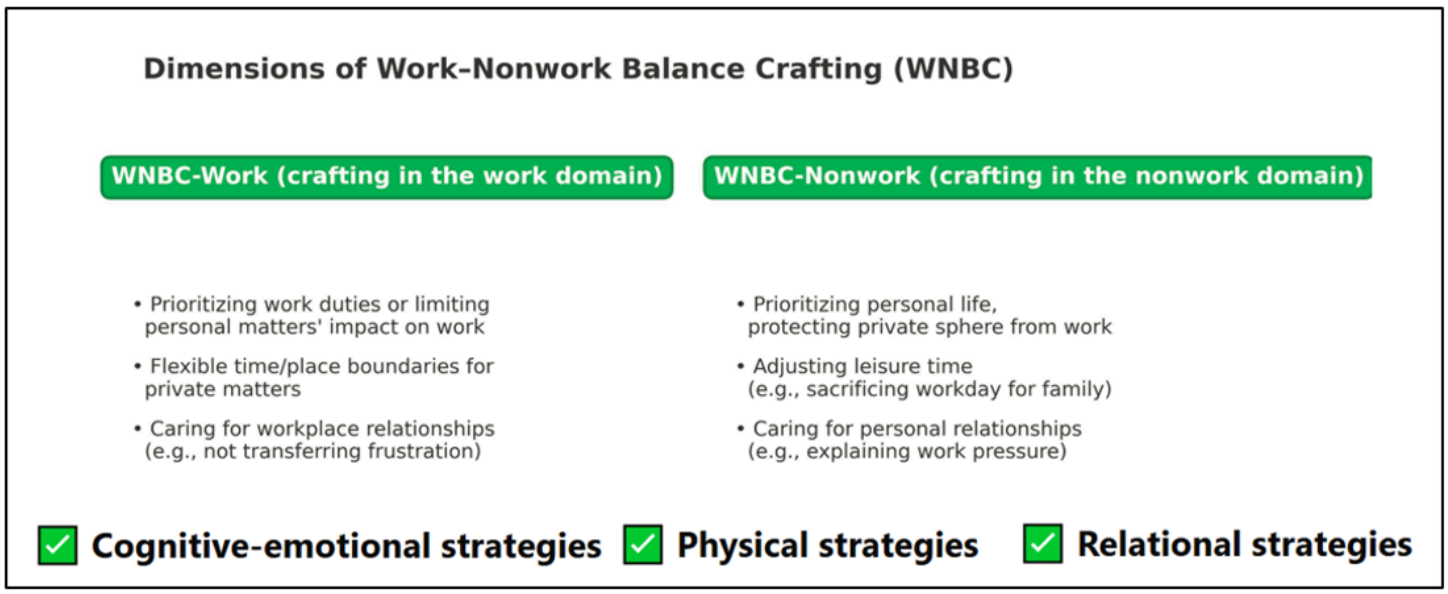


Figure 1

Scheme of the dimensions of Work-Nonwork Balance Crafting (WNBC) and the kinds of individual job-crafting strategies

Source: own study based on Kerksieck et al. (2022).

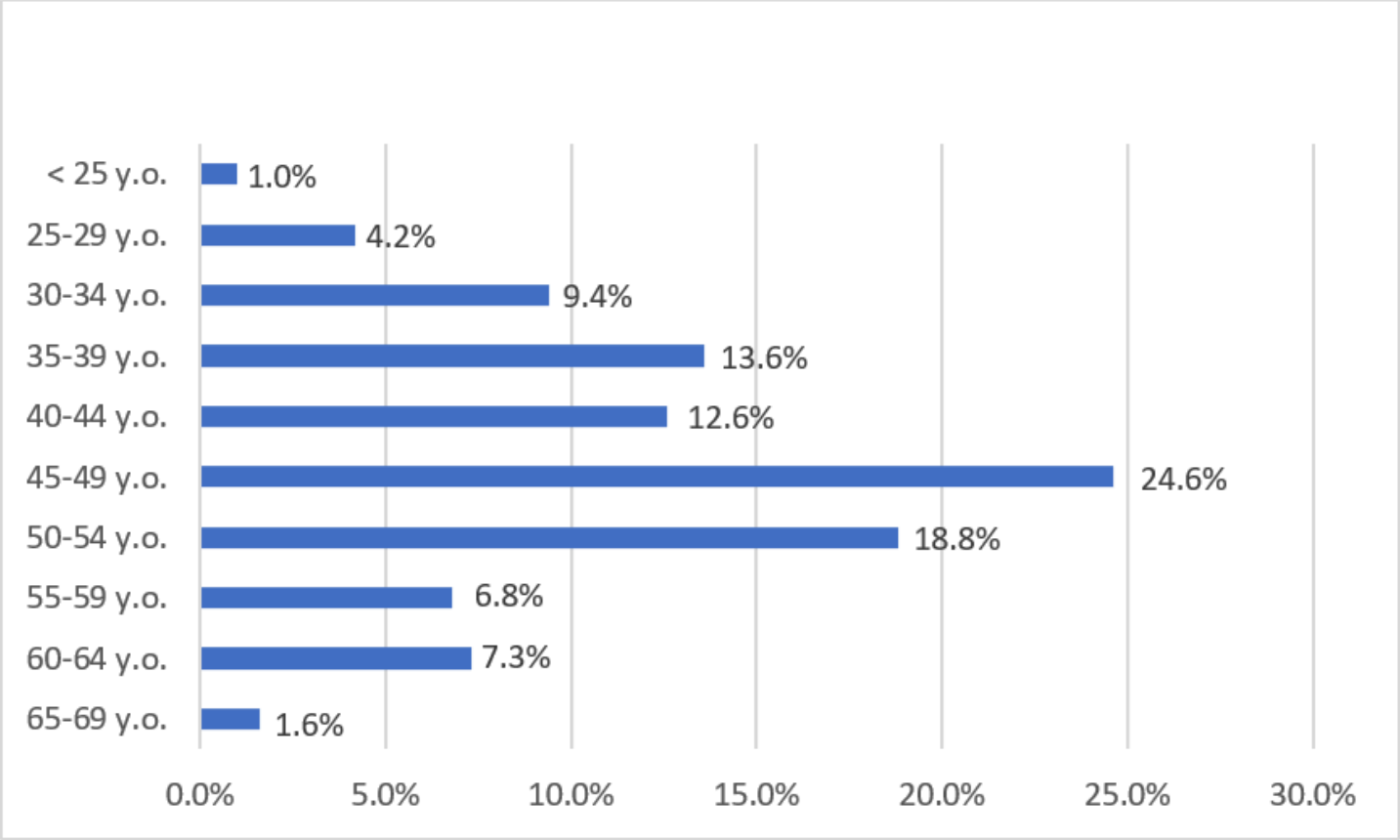


Figure 2

Respondent structure by age

Source: own study.

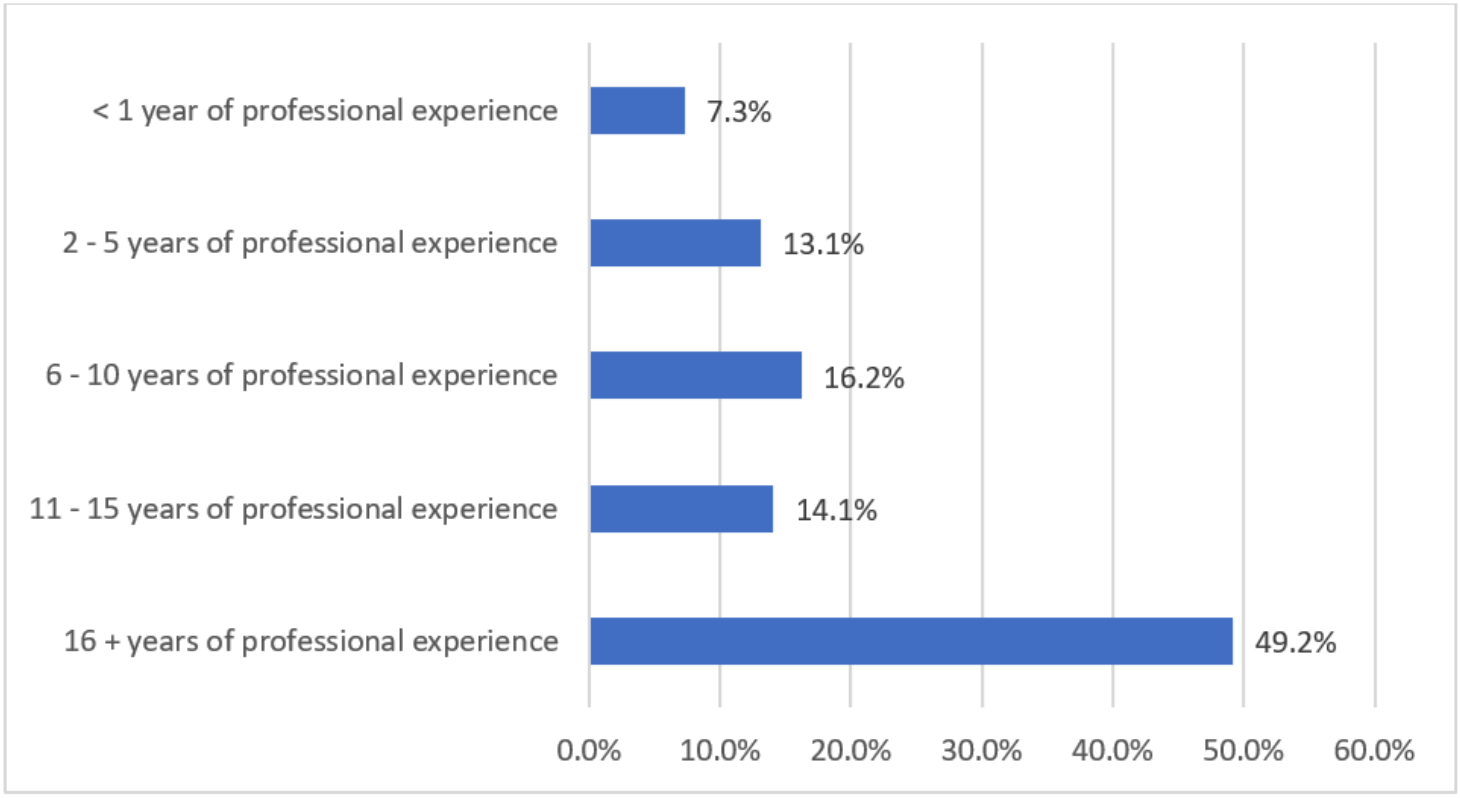


Figure 3

Respondent structure by professional experience

Source: own study.

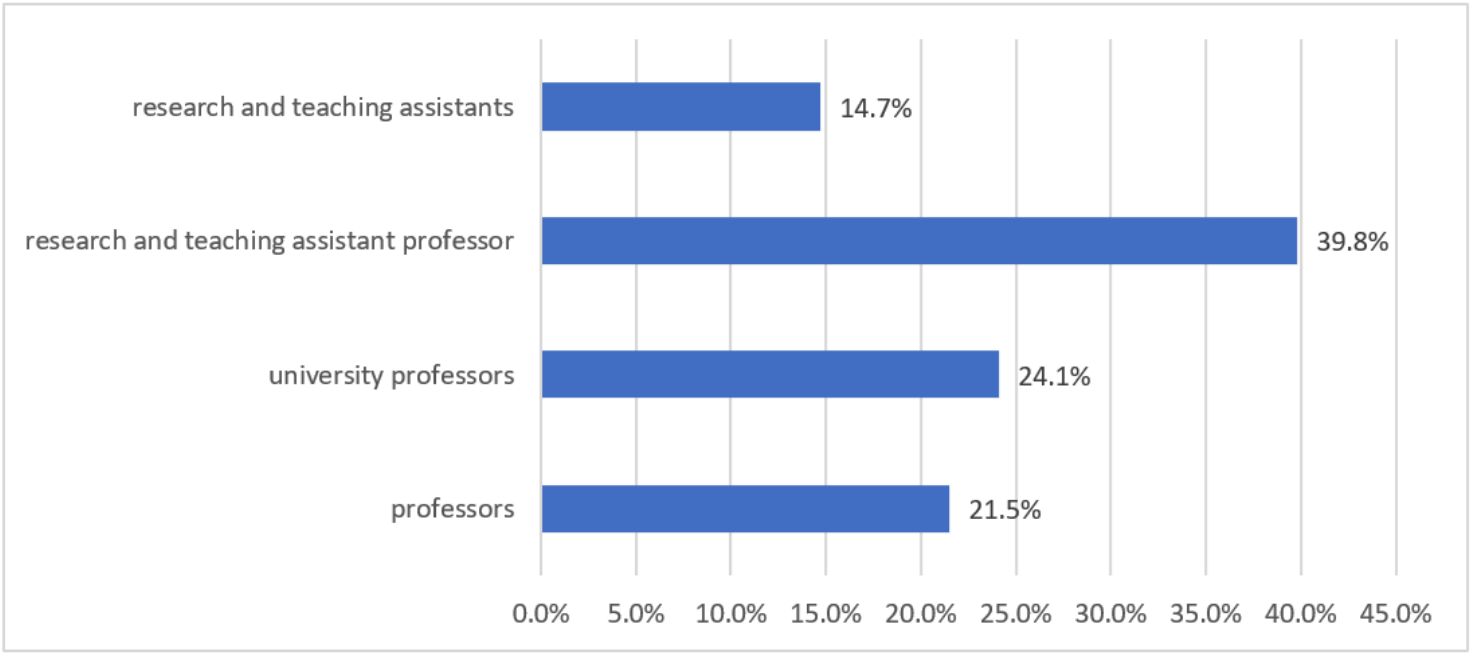


Figure 4

Respondent structure by position held

Source: own study.

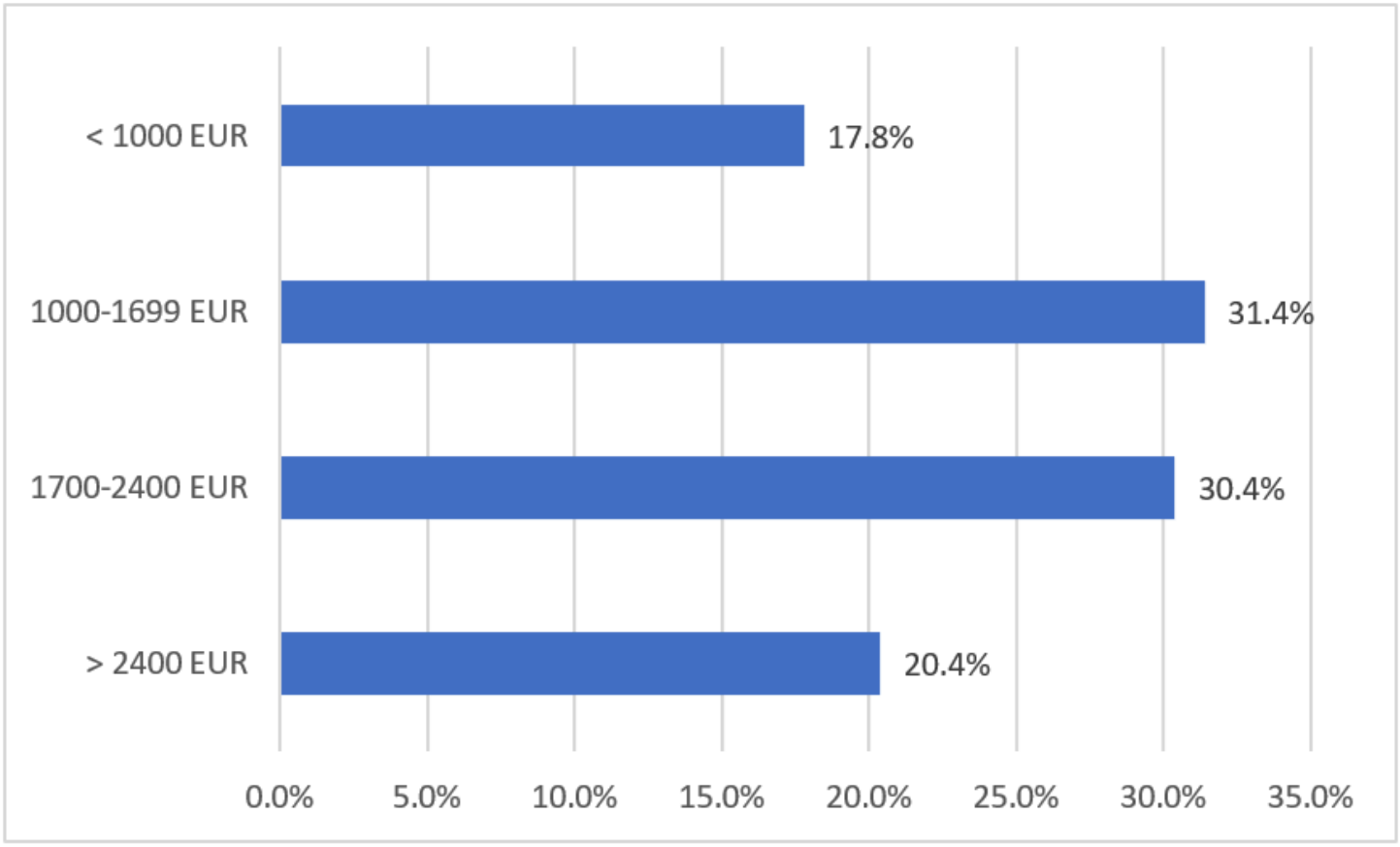


Figure 5

Respondent structure by salary (in EUR)

Source: own study.

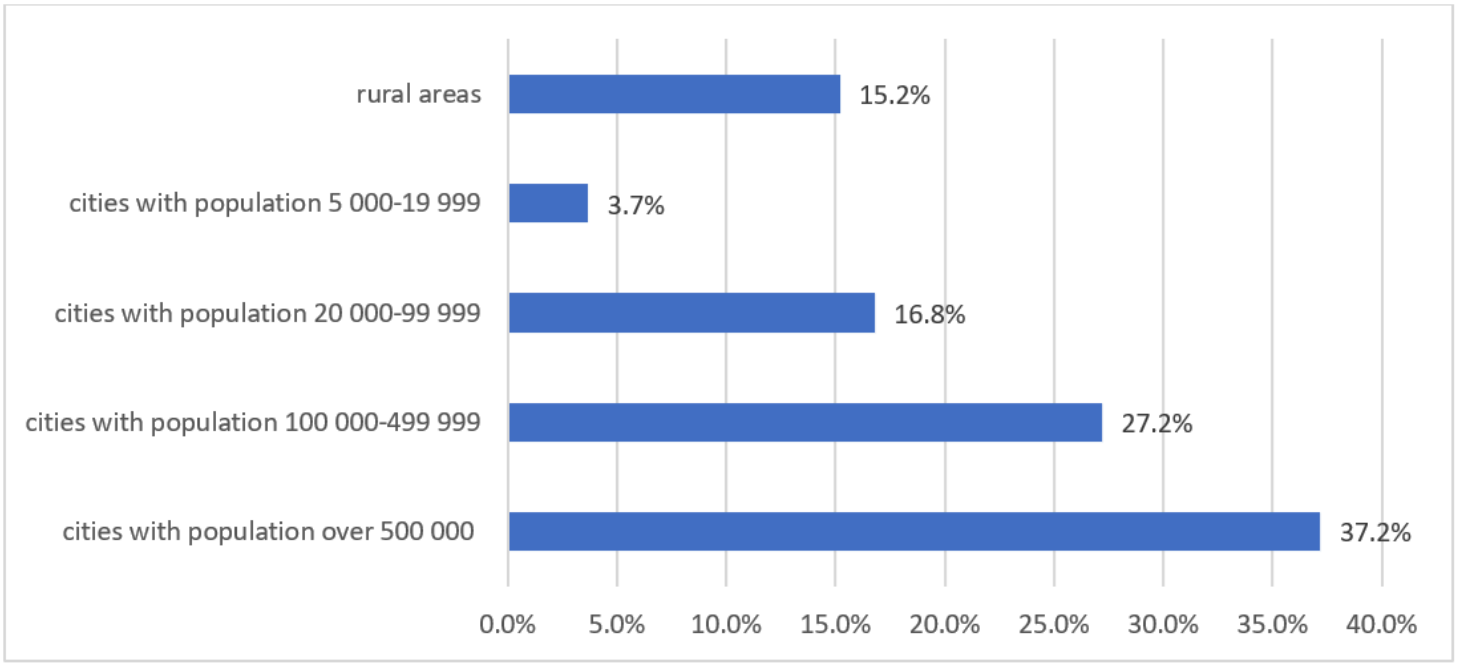


Figure 6

Respondent structure by class of place of residence

Source: own study.

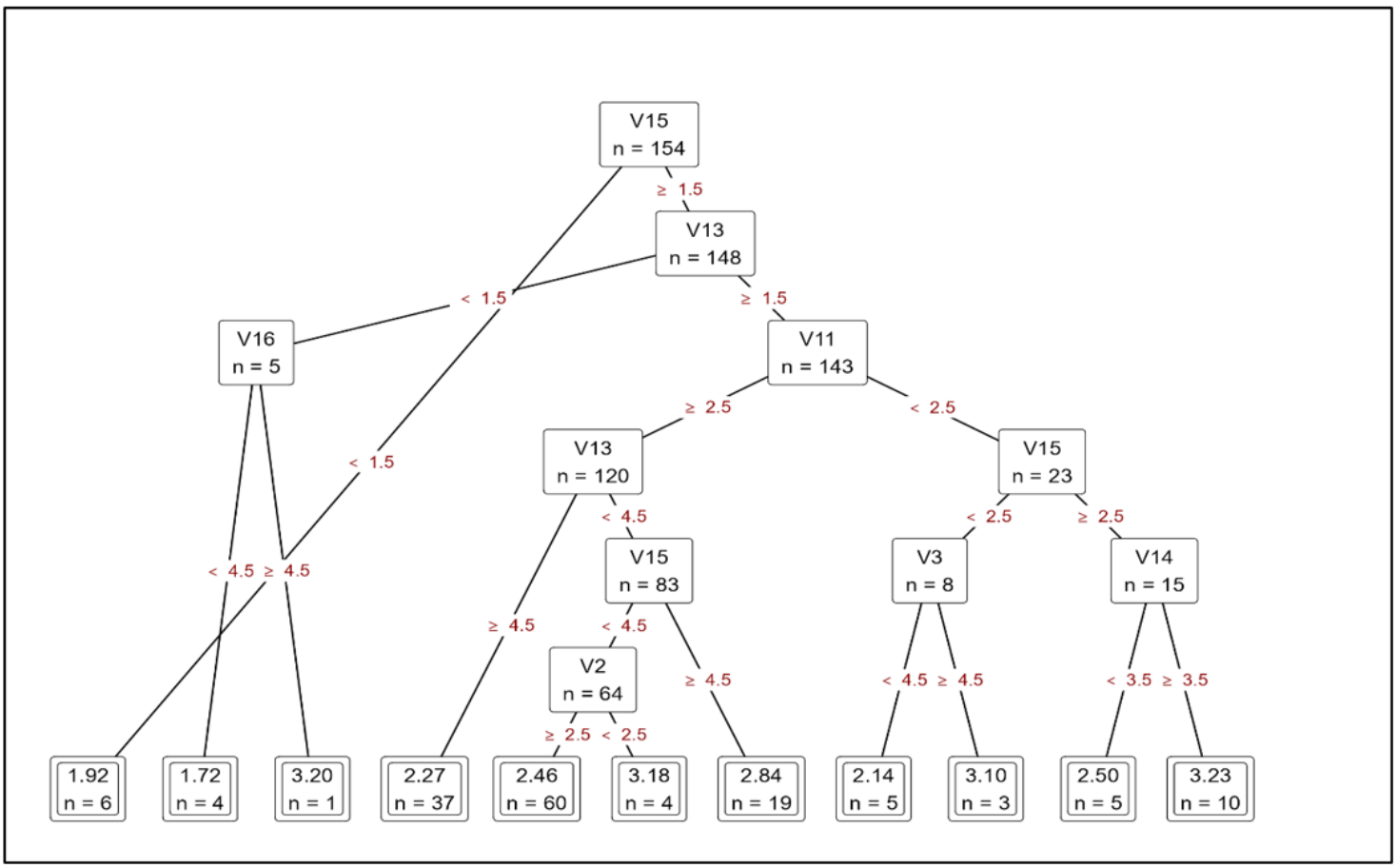


Figure 7

Structure of the C&RT decision tree predicting the level of professional digitalization based on variables from the Work–Nonwork Balance Crafting scale

Source: own study based on the analysis performed with the JASP program (2025).

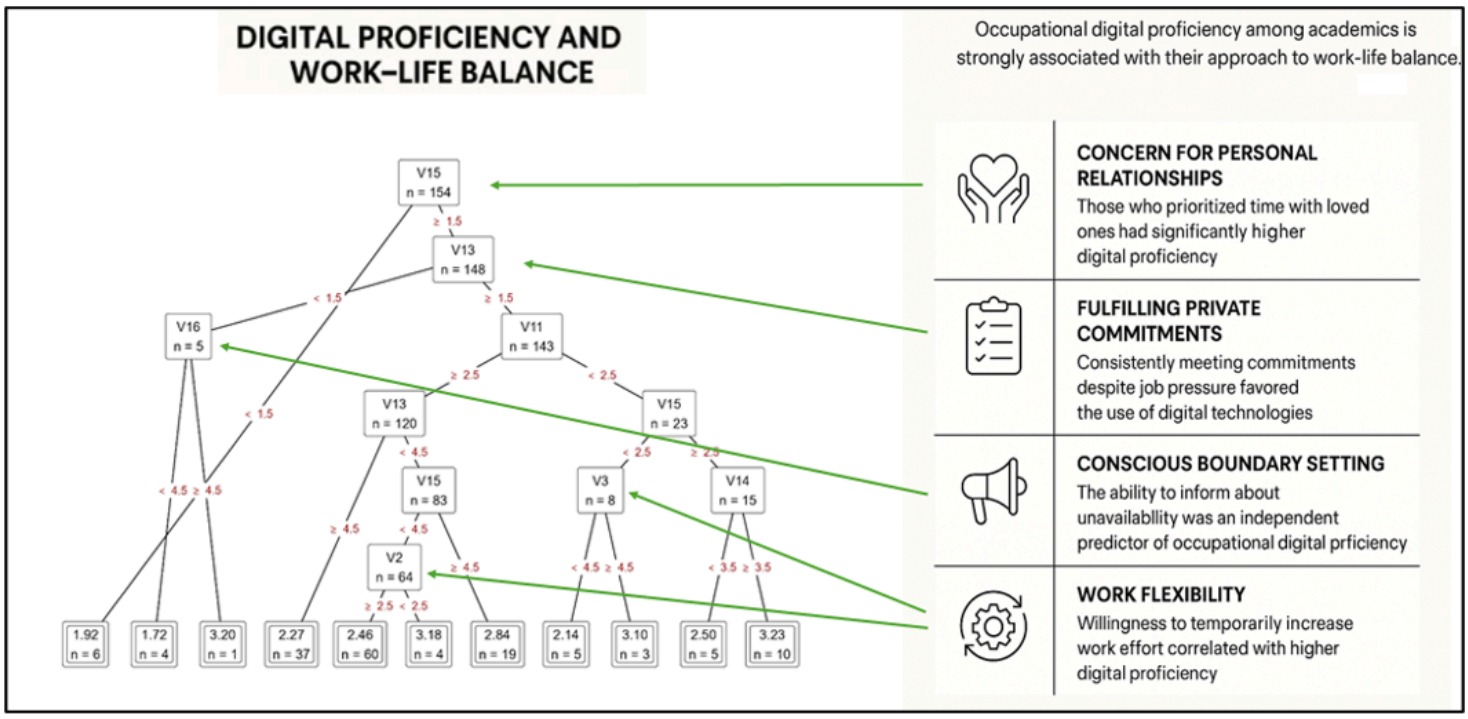


Figure 8

The work–life balance (WLB) factors identified as the most significant in differentiating the level of occupational digitalization among university teachers.

Source: own study.

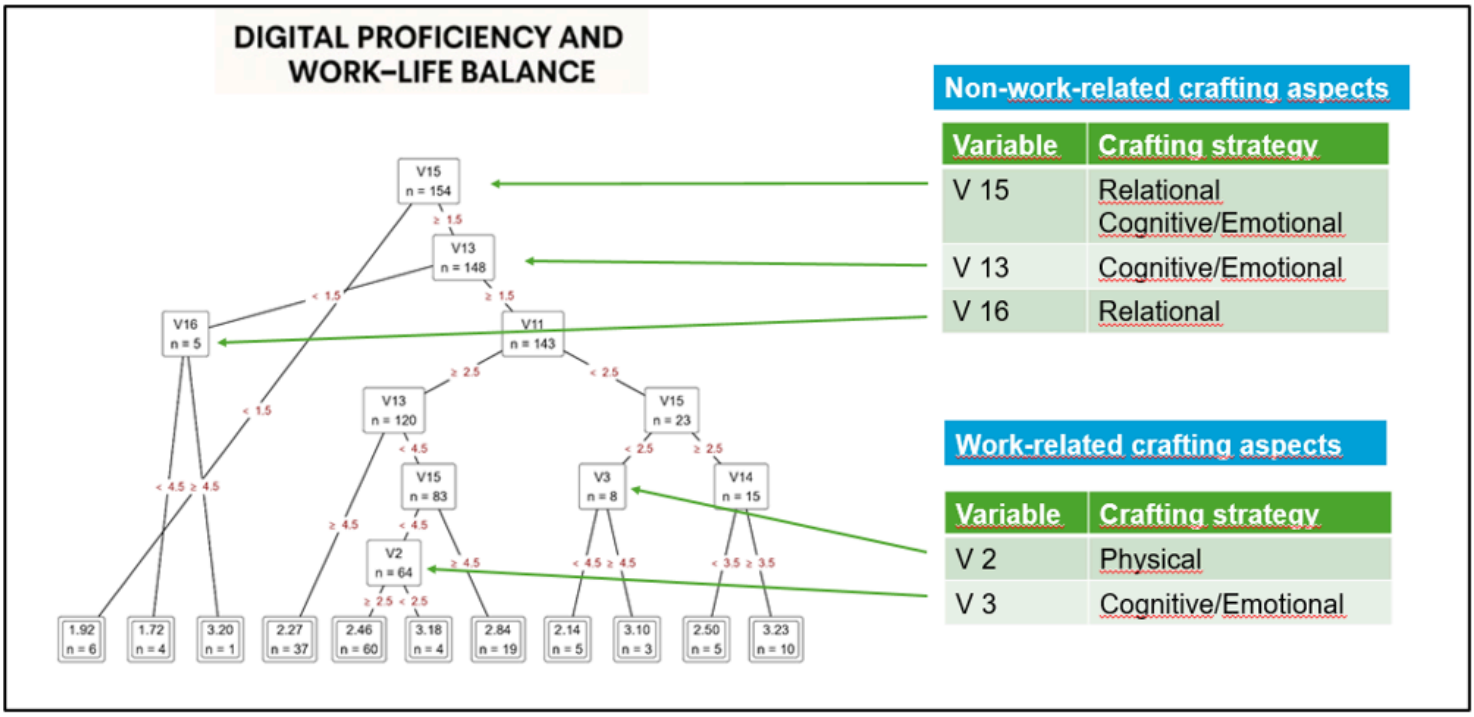


Figure 9

WLB-related attitudes and strategies identified as the most significant predictors of a high level of digital technology use in teaching and research activities

Source: own study.